

Problem Set 4

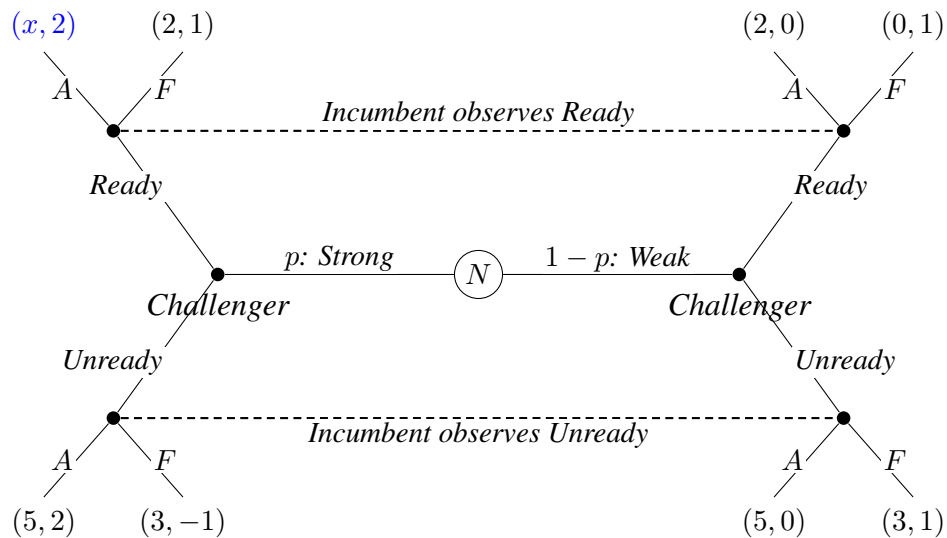
POLS W4732

Fall 2026

Due November 6 before the TA section

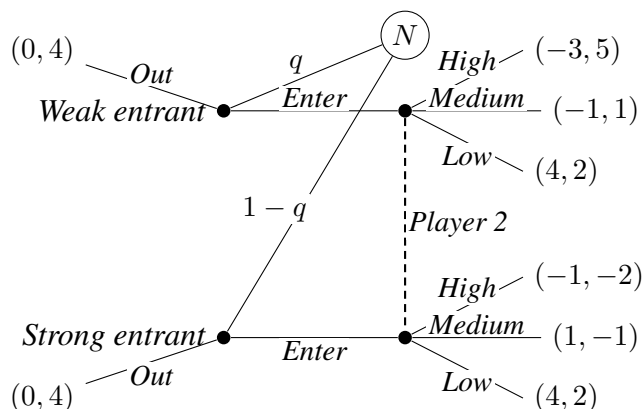
Artificial-intelligence policy. Unless an exercise explicitly states otherwise, students may not use generative-AI systems (including ChatGPT, Claude, Gemini, Copilot, or AI-generated search answers) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, typesetting software, the teaching team, and collaboration within the registered problem-set group are permitted. Any approved computational tool must be disclosed. Every group member is responsible for understanding and being able to explain the entire submission.

Exercise 1 (Crisis bargaining). *Nature draws a Strong Challenger with probability p and a Weak Challenger with probability $1-p$. The Challenger observes its type and chooses Ready or Unready. The Incumbent observes that action but not the type and chooses A or F. Payoffs are shown as (Challenger, Incumbent).*



The Strong Challenger's payoff at (Strong, Ready, A), originally 4, is the parameter x . For each candidate separating profile, derive both types' incentive constraints, the Incumbent's sequential-rationality conditions, and Bayes-consistent beliefs. Characterize all x for which a separating PBE exists and identify the first condition to fail at the boundary.

Exercise 2 (Informed entry). Nature draws a Weak entrant with probability q and a Strong entrant with probability $1 - q$. The entrant observes its type and chooses Out or Enter. After Enter, Player 2 cannot observe type and chooses High, Medium, or Low.



For each profile below, supply all required on- and off-path beliefs and classify it as impossible for every belief, sustainable for some off-path belief, or sustainable only for a unique belief. Give the relevant inequality.

1. Both types choose Out; Player 2 chooses High after Enter.
2. Both types choose Enter; Player 2 chooses Low.
3. Strong chooses Enter, Weak chooses Out; Player 2 chooses Low.

Then characterize, as a function of q , when each pooling profile can be supported and how Player 2's action changes at the threshold.

Exercise 3 (Spence signaling). A worker has type $\theta \in \{1, 2\}$ and productivity $v_\theta = \theta$. The prior probability of type 1 is π . A competitive market pays expected productivity conditional on signal $z \geq 0$. Type 1 pays $C_1(z) = z$ and type 2 pays $C_2(z) = z(1 - \epsilon)$, where $\epsilon \in (0, 1)$.

1. Derive the separating-signal interval from both incentive constraints, including endpoint tie-breaking.
2. For a proposed pooling signal z_p , construct the least and most favorable off-path wage beliefs for sustaining pooling. Determine the pooling set under each restriction.
3. Compare type payoffs, distinguishing conclusions true in every PBE from those true only after equilibrium selection.
4. Replace the high type's cost by $z(1 - \epsilon) + az^2$, $a > 0$. Determine which parts of the single-crossing argument survive.